

Reg No.:

Name:

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S2 (R) Examination May 2025 (2024 Scheme)

Course Code: PCITT205

Course Name - DISCRETE MATHEMATICAL STRUCTURES

Max. Marks: 60

Duration: 2 hours 30 minutes

PART A

(Answer all questions. Each question carries 3 marks)

		CO	Marks
✓	What are the laws of logical equivalences?	CO1	(3)
✓	Negate quantifiers. Explain with examples.	CO1	(3)
✓	Find the least number of cables required to connect eight computers to four printers to guarantee that for every choice of four of the eight computers, these four computers can directly access four different printers. Justify your answer using pigeon hole principle.	CO2	(3)
✓	What is the coefficient of $x^{101}y^{99}$ in the expansion of $(2x - 3y)^{200}$?	CO2	(3)
5	Define a partial ordering. Show that the divisibility relation on the set of positive integers is a partial order.	CO3	(3)
6	Define the binary operations $\oplus$ and $\odot$ on $\mathbb{Z}$ by $x \oplus y = x + y - 7$, $x \odot y = x + y - 3$, for any $x, y \in \mathbb{Z}$. Explain why $(\mathbb{Z}, \oplus, \odot)$ is not a ring.	CO3	(3)
✓	What is exponential generating function? Give example.	CO4	(3)
✓	Solve the recurrence relation $6a_n - 7a_{n-1} = 0$, $n \geq 1$ and $a_3 = 343$.	CO4	(3)

PART B

(Answer any one full question from each module, each question carries 9 marks)

Module -1

- 9/ a) Obtain the disjunctive normal form $p \rightarrow (p \rightarrow q) \wedge \sim (\sim q \vee \sim p)$. CO1 (3)
- b) Write the rules of inference. CO1 (6)
- 10/ a) Construct truth table for $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$. CO1 (3)
- b) Express each of these statements using logical operators, predicates, and quantifiers. CO1 (3)
- i. Some propositions are tautologies.

- ii. The negation of a contradiction is a tautology.
 iii. The disjunction of two contingencies can be a tautology.
 iv. The conjunction of two tautologies is a tautology.
 c) Show that the implication $((p \vee q) \wedge (\sim p \vee r)) \rightarrow (q \vee r)$ is a tautology. CO1 (3)

Module -2

- 11 a) There are 2504 computer science students at a school. Of these, 1876 have taken a course in Java, 999 have taken a course in Python, and 345 have taken a course in C. Further, 876 have taken courses in both Java and Python, 231 have taken courses in both Python and C, and 290 have taken courses in both Java and C. If 189 of these students have taken courses in Python, Java, and C, how many of these 2504 students have not taken a course in any of these three programming languages? CO2 (5)
 b) How many arrangements are there of all the letters in **SOCIOLÓGICAL?** CO2 (2)

In how many of above arrangements are A and G adjacent?

In how many of above arrangements are all the vowels adjacent?

- c) Write the rule of sums and product. Give examples. CO2 (2)
 12 a) Determine the number of numbers less than 500 and are not divisible by 2,3,5,6,10. CO2 (5)
 b) How many different paths in the x y-plane are there from (0,0) to (7,7) if a path proceeds one step at a time by going either one space to the right (R) or one space upward (U)? CO2 (4)
 How many such paths are there from (2,7) to (9,14)?
 Give a general formula for these two results.

Module -3

- 13 a) Let $A = \{1,2,3,4\}$ and $B = \{x, y, z\}$ CO3 (5)
 i. List five functions from A to B.
 ii. How many functions $f: A \rightarrow B$ are there?
 iii. How many functions $f: A \rightarrow B$ are one-to-one?
 iv. How many functions $g: B \rightarrow A$ are there?
 v. How many functions $g: B \rightarrow A$ are one-to-one?
 b) Define an equivalence relation. Show that the relation $R = \{(a, b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation whenever m is a positive integer. CO3 (4)

- 14 a) Draw the Hasse diagram for the divisibility relation on $\{2,4,20,25\}$ starting from the digraph. CO3 (3)
 b) Let $f, g, h: \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(x)=x-1$, $g(x)=3x$, $h(x) = \begin{cases} 0, & x \text{ even} \\ 1, & x \text{ odd} \end{cases}$ CO3 (6)
 Determine: $f \circ g$, $g \circ f$, $g \circ h$, $h \circ g$, $f \circ (g \circ h)$ and $(f \circ g) \circ h$.

Module -4

- 15 a) Find the generating function for each of these sequences that follows the pattern of the initial terms listed. CO4 (5)
 i. 0, 2, 2, 2, 2, 2, 0, 0, 0, 0, ...
 ii. 0, 0, 0, 1, 1, 1, 1, 1, ...
 iii. 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, ...
 iv. 2, 4, 8, 16, 32, 64, 128, 256, ...
 v. 2, -2, 2, -2, 2, -2, 2, -2, ...
 b) Solve these recurrence relations together with the initial conditions given. CO4 (4)
 $a_n = 7a_{n-1} - 10a_{n-2}$ for $n \geq 2$, $a_0 = 2$, $a_1 = 1$
 16 a) Use generating functions to find the number of ways to select 14 balls from a jar containing 100 red balls, 100 blue balls, and 100 green balls so that no fewer than 3 and no more than 10 blue balls are selected. Assume that the order in which the balls are drawn does not matter. CO4 (4)
 b) Solve the recurrence relation: $a_n = 2a_{n-1} + 2^n$, $a_0 = 2$. CO4 (5)
